OENG1209: Engineering Science

Week 6:

Work & Energy

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Review and motivation

MECHANICS: Position → Velocity → Acceleration ∝ Force

Kinematic equations:

$$\vec{r} \Rightarrow \vec{v} = \frac{d\vec{r}}{dt} \Rightarrow \vec{a} = \frac{d\vec{v}}{dt}$$

position

Newton's second law:

$$\vec{F}_{net} = m\vec{a}$$

Observations:

These equations involve **VECTOR** quantities → Their **DIRECTIONS**

An alternative:

prove base on Newton's law

→ Using the **ENERGY** approach

don't care about direction

WHY?: Only need to deal with **SCALAR** quantities.

Today's Topics

1. Work
2. Kinetic Energy & the Kinetic Energy – Work theorem
3. Potential Energy
4. Energy Conservation

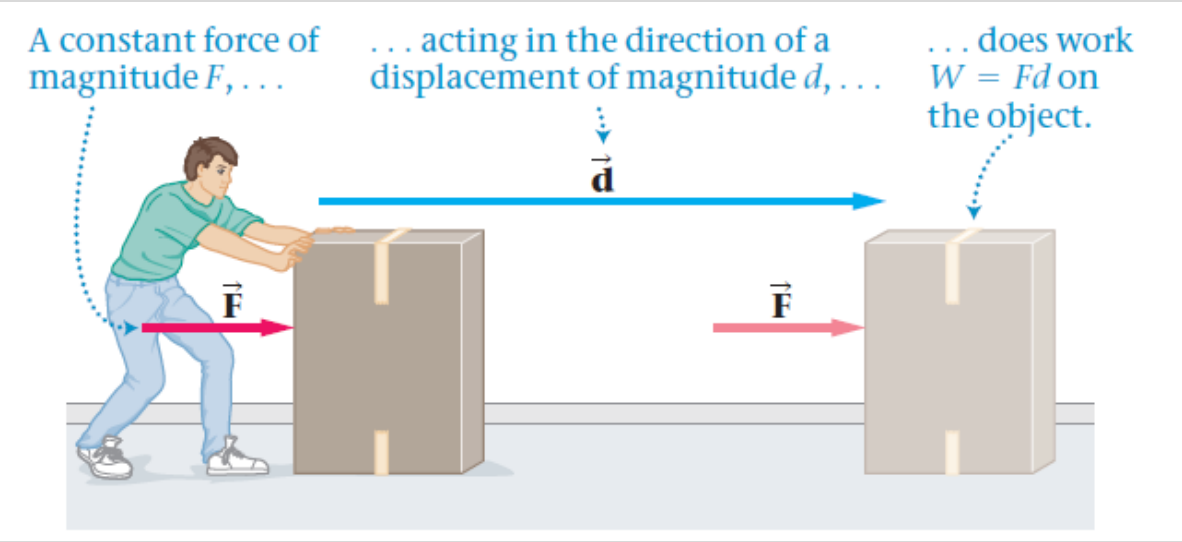
1. Work

The thing that keeps the world moving

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Work
done by a constant force

Suppose that $F = const$ and it is applied in the direction of Displacement.



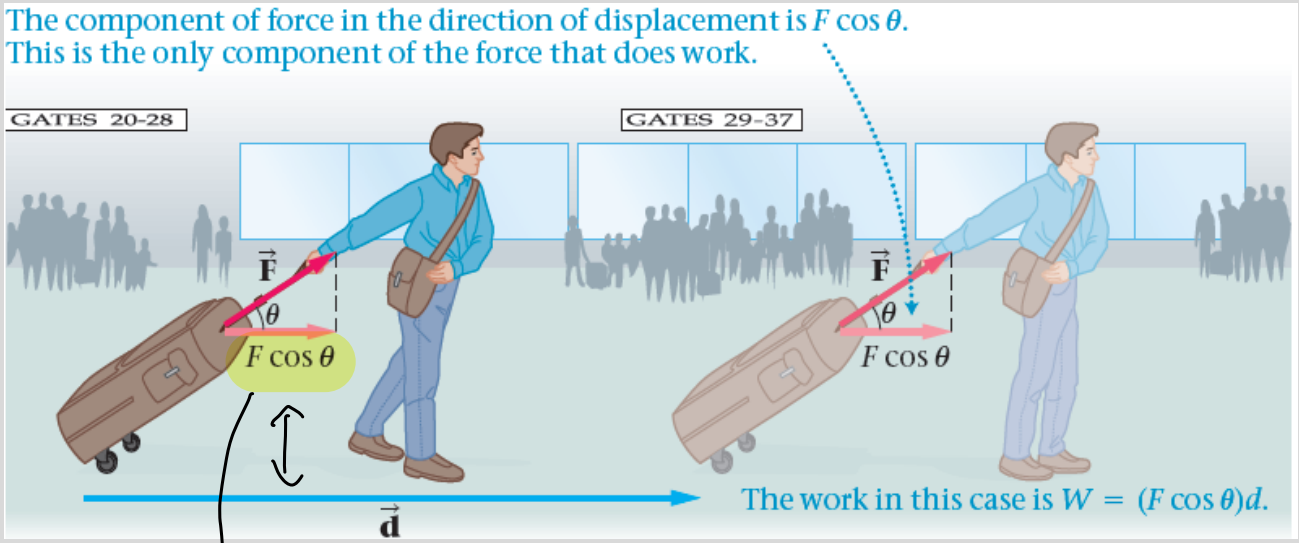
SI Unit: newton-meter (N.m) → J, joule. $F \cdot d = W$ $\vec{F}$ same direction with $\vec{d}$

Work is a **SCALAR** quantity.

↑ the performance

Work

done by a force at an angle



Same direction with $\vec{d}$

measure the performance in either vertical, or horizontal

$\cos 0 = 1$

best peform at it

Work
Negative work

By its definition:

IF $\vec{F} \uparrow\downarrow \vec{d} \rightarrow \theta = 180^\circ \Rightarrow \cos\theta = -1$

$$W = Fd\cos\alpha$$

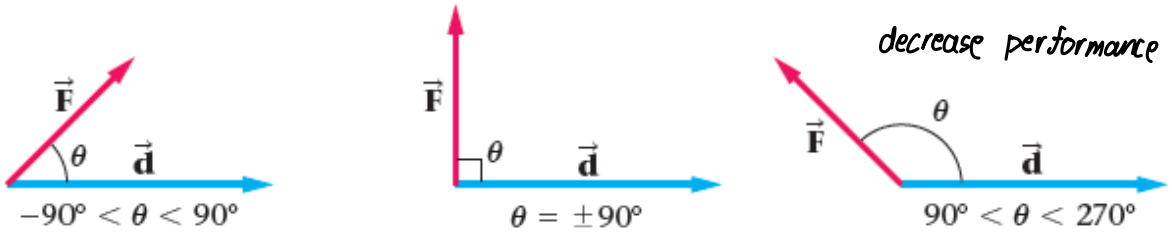
(i) *Work is positive if the force has a component in the direction of motion* $(-90° < \theta < 90°)$.

(ii) *Work is zero if the force has no component in the direction of motion* $(\theta = \pm 90°)$.

(iii) *Work is negative if the force has a component opposite to the direction of motion* $(90° < \theta < 270°)$.

Thus, whenever we calculate work, we must be careful about its sign and not just assume it is positive.

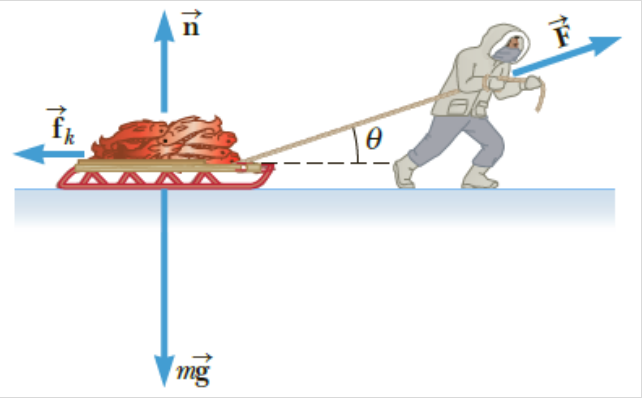
decrease performance



(a) Positive work **(b)** Zero work **(c)** Negative work

Work
Example

A man returning from a successful ice fishing trip pulls a sled loaded with salmon. The total mass of the sled and salmon is 50 kg, and the man exerts a force of maginitude 120 N on the sled by pulling on the rope. (a) If $\theta = 0$, compute the work if he pulls the sled 5 m? (b) Compute the work if $\theta = 30^o$

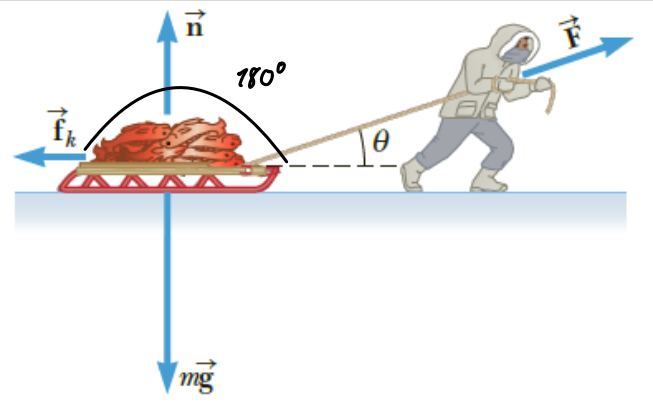


a) $\theta = 0 \rightarrow W = Fd = (120N)(5m) = 600J$

b) $\theta = 30° \rightarrow W = \cos 30° (120N)(5m) = 520J$

Work
Example

... (c) At a coordinate position of 12.4 m, the man lets up on the applied force. A friction force of 45 N between the ice and the sled brings the sled to rest at a coordinate position of 18.2 m. How much does friction do on the sled?



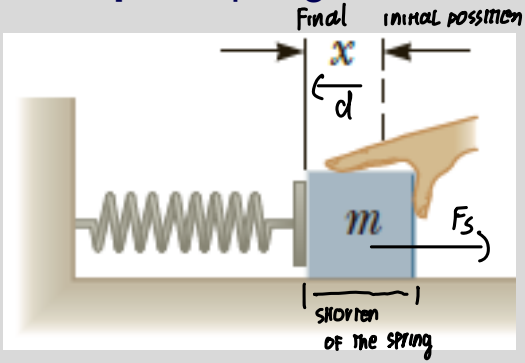
$d = x_f - x_i = (18.2) - (12.4) = 5.8\ m$

$\theta = 180^\circ \Rightarrow \cos 180 = -1$

$W = -1(45)(5.8) = -200\ J$

Work
done by a varying force

Example: Spring force



WHAT IS the compression in the spring

$F_s = K \cdot x$

F_s — F_s

F_s vs. x: K -> slope; the F_s is too small (neglect); $\frac{1}{2} kd \cdot d = \frac{1}{2} kd^2$

F vs. d: $W = F \cdot d$

W = area under the curve F vs d

$W = -\frac{1}{2} kd \cdot d = -\frac{1}{2} kd^2$

opposite direction with force

integral $\int F(x)$

2. Kinetic Energy & the Kinetic Energy – Work theorem

Energy associated with MOTION

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Kinetic Energy

Using Newton's second law can be difficult if the forces involved are complicated

→ It is easier to use the work due to the net force.

Newton's 2nd law: $F_{net} = ma$

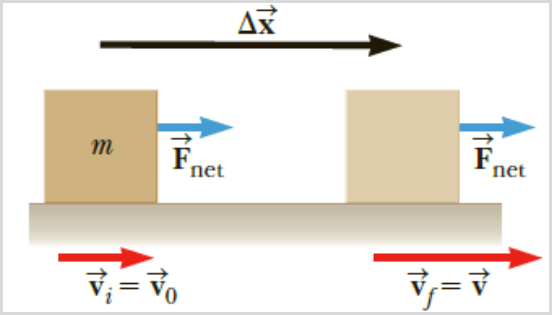
keep F_{net} constant → a = const

$$v^2 = v_0^2 + 2a\Delta x$$

$F_{net} \cdot \Delta x = W$

$\therefore \underbrace{F_{net} \cdot \Delta x}_{W_{net}} = ma\Delta x = m\frac{1}{2}(v^2 - v_0^2)$

$= \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$



$$W_{net} = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

The Kinetic Energy – Work theorem

$$W_{net} = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

The net work done on an object equals a change in quantity of the form $mv^2/2$.

→ involves speed v → interpreted as energy associated with **motion**

→ **KINETIC** energy

The **kinetic energy** ***KE*** of an object of mass m moving with a speed v is

$$KE \equiv \tfrac{1}{2}mv^2$$

SI unit: joule (J) = kg · m²/s²

The net work done on an object is equal to the change in the object's kinetic energy:

$$W_{\text{net}} = KE_f - KE_i = \Delta KE$$

where the change in the kinetic energy is due entirely to the object's change in speed.

The Kinetic Energy – Work theorem

Example: A 220-kg motorcycle is cruising at 14 m/s. What is the net work that must be done on the motorcycle to increase its speed to 19 m/s?

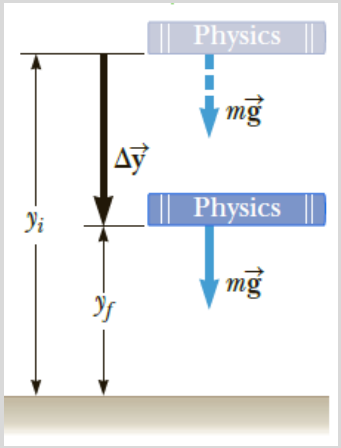
$$W_{net} = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = 18\,000\,J$$

Conservative and Nonconservative forces

Definition of Conservative force:

A force is conservative if the work it does moving an object between two points is the same **no matter what path is taken.**

Example: Gravitational force

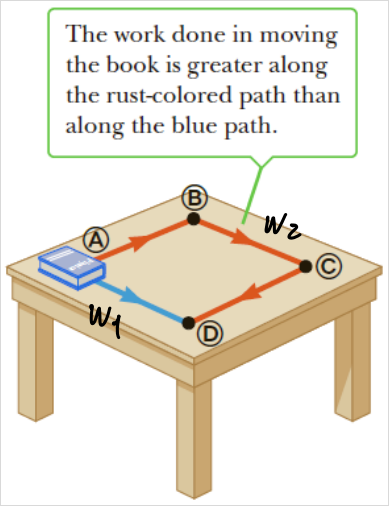


Work = $\Delta y \cdot -1 \cdot W$ / weight

$(y_f - y_i)$

Otherwise →**NONconservative** force.

Example: Friction force



$W_1 < W_2$

↑ longer distance

$W_{net} = W_{nc} + W_c = \Delta KE$

↑ not conservative

↑ con...

very confuse

3. Potential Energy

Energy associated with relative POSITION

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Potential Energy

Gravitational potential energy

Consider a book of mass m falls from a height y_i to a height y_f,

the work-energy theorem can be written as

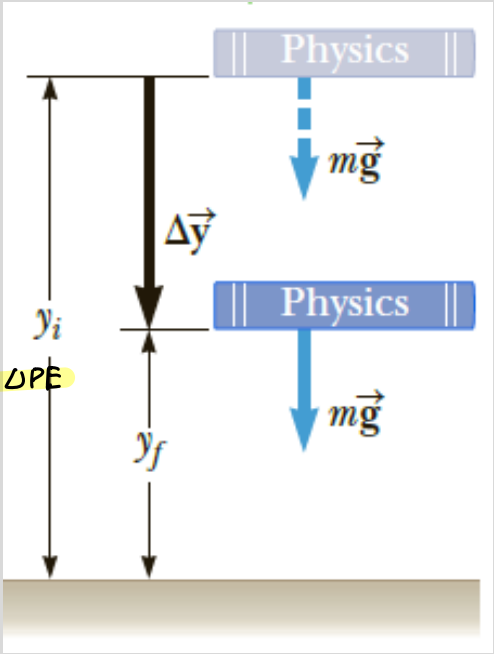
$$W_{net} = \Delta KE$$

$W_{net} = W_{nc} + W_c$

W_c ↑ Weight $\Delta y = -mg(y_f - y_i)$

$\Rightarrow W_{nc} = \Delta KE + mgy_f - mgy_i = \Delta KE + PE_f - PE_i = \Delta KE + \Delta PE$

We define: $PE = mgy$ → gravitational potential energy.



$$W_{nc} = \Delta KE + \Delta PE$$

Potential Energy

Gravitational potential energy

$$W_{nc} = \Delta KE + \Delta PE = 0$$

Conservative systems:

$$W_{nc} = 0 \Rightarrow$$

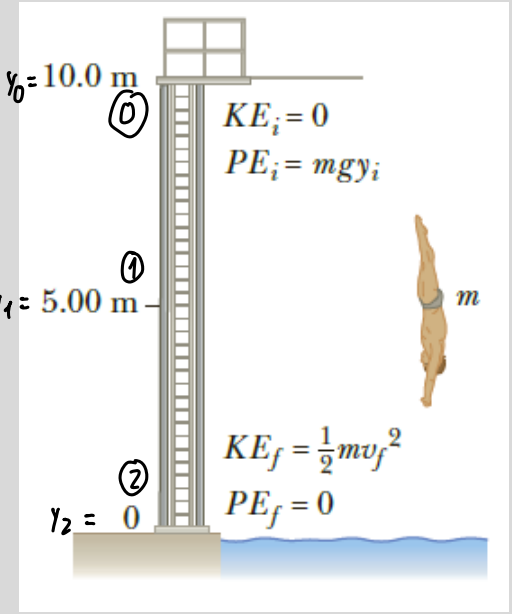
$$KE_f - KE_i + PE_f - PE_i = 0$$

$$KE_i + PE_i = KE_f + PE_f$$

Potential Energy

Gravitational potential energy

Example: A diver of mass m drops from a board 10 m above the water's surface. Neglect air resistantce. (a) Find his speed 5.00 m above the water's surface. (b) Find his speed as he hits the water.



$0 \to 1 \Rightarrow \cancel{KE_0}^{0} + PE_0 = KE_1 + PE_1$

$\Rightarrow \cancel{mg}\, y_0 = \frac{1}{2}\cancel{m}v_1^2 + \cancel{mg}\, y_1$

$v_1 = \sqrt{2g(y_0 - y_1)} = 9.9\ m/s$

$0 \to 2 \Rightarrow \cancel{KE_0}^{0} + PE_0 = KE_2 + \cancel{PE_2}^{0}$

$mg\, y_0 = \frac{1}{2}mv_2^2 \Rightarrow v_2 \sqrt{2gy_0} = 14\ m/s$

Potential Energy

Gravitational potential energy

$$W_{nc} = \Delta KE + \Delta PE$$

Non-conservative systems:

$$W_{nc} \neq 0 \Rightarrow$$

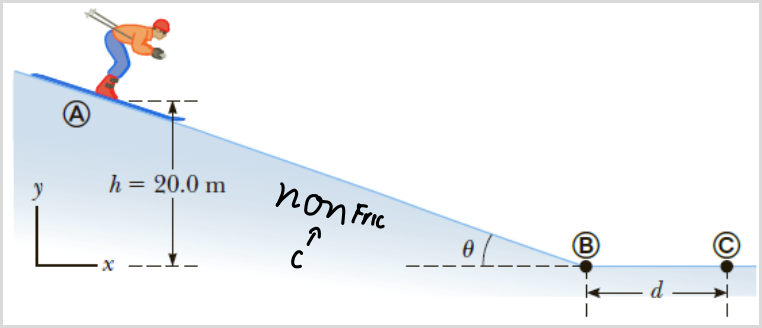
$W_{nc} = KE_f - KE_i + PE_f - PE_i$

$\Rightarrow KE_f + PE_f = KE_i + PE_i + W_{nc}$

Potential Energy

Gravitational potential energy

Example: A skier starts from rest at the top of a frictionless incline of height 20 m. At the bottom of the incline, the skier encouters a horizontal surface where the coefficient of kinetic friction between skis and snow $\mu_k = 0.21$. (a) Find the skier's speed at the bottom. (b) How far does the skier travel on the horizontal surface before coming to rest? Neglect air resistance.



Ⓐ→Ⓑ $v_A = 0$: $KE_A + PE_A = KE_B + PE_B$

$mgh = \frac{1}{2}mv_B^2$

$v_B = \sqrt{2gh}$

Ⓑ→Ⓒ $KE_B + PE_B + W_{nc} = KE_C + PE_C$

$\frac{1}{2}mv_B^2 - F_k \cdot d = \frac{1}{2}mv_C^2 = 0$

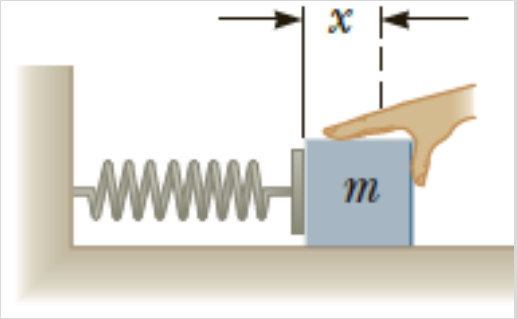
$\frac{1}{2}mv_B^2 - \mu_k mgd = 0$

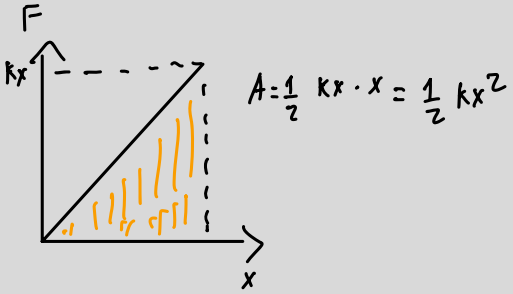
$\Rightarrow d = \frac{v_B^2}{2\mu_k g}$

Potential Energy
Spring Potential Energy

$$W_{net} = \Delta KE$$

The work done by the spring force $W_s = ?$ if the shorten of the spring is $x = x_f - x_i$





We define the spring potential energy as

$$PE_s = \frac{1}{2}kx^2$$

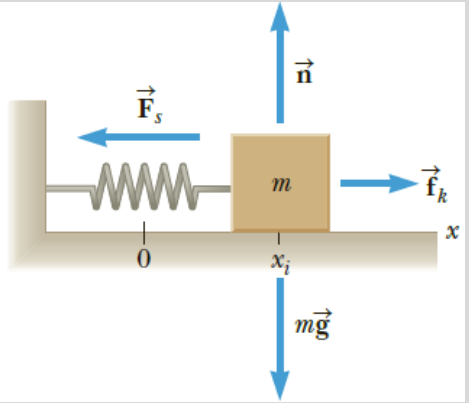
Now,

Potential Energy
Spring Potential Energy

Example: The surface is **fricitionless.** The block is pulled out to $x_i = 0.05m$ and released. Find the speed of the block when

a) it first reaches the equilibrium point.

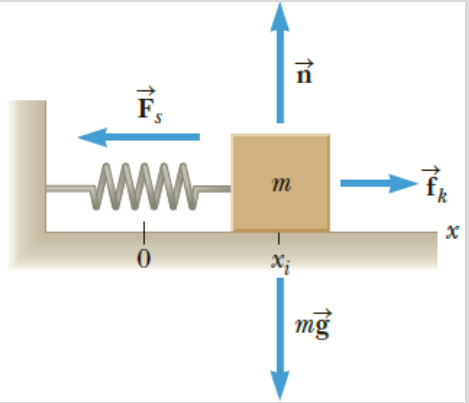
b) x = 0.025 m.



Potential Energy
Spring Potential Energy

Example: Repeat part (a) if friction acts on the block, with $\mu_k = 0.15$.

a) it first reaches the equilibrium point.



4. Energy conservation

Energy of ISOLATED systems is constant

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Energy conservation

The principle of the conservation of energy states that

- Energy can't be created or destroyed.
- It can be transformed, bu the total energy content of any ISOLATED system is always constant.

Recall that the work-energy theorem can be written as

$$W_{nc} + W_c = \Delta KE$$

We define the total MECHANICAL energy $E = KE + PE$

So, **the work done on a system by all nonservative forces is equal to the change in mechanical energy of that system**

$$W_{nc} = E_f - E_i = \Delta E$$